CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 22: Data Handling

Topic-wise short notes with solved examples

Organising raw data into frequency tables, grouped data and class intervals, and showing data with bar graphs, histograms and pie charts.

1 Data

- **Data** are numerical facts collected for a purpose. Data as first collected are **raw data**.
- Each entry is an **observation**. The number of times an observation occurs is its **frequency**.
- **Range** = highest observation − lowest observation.
- **Tally marks** make counting easy: four upright strokes and a fifth across them make a bundle of five.
- A **frequency distribution table** lists the observations with their frequencies.

Example 1.1 The marks of 20 students are: 5, 7, 6, 8, 5, 9, 7, 6, 7, 8, 5, 7, 9, 6, 7, 8, 10, 7, 6, 8. Make a frequency table.

SOLUTION

Count each mark: 5 occurs 3 times, 6 four times, 7 six times, 8 four times, 9 twice, 10 once.

Frequencies 3, 4, 6, 4, 2, 1 (total 20)

Marks	5	6	7	8	9	10
Frequency	3	4	6	4	2	1

2 Grouped data

- When there are many different observations, they are put into groups called **class intervals**.
- **Class limits**: in the class 10–20, the lower limit is 10 and the upper limit is 20.
- **Exclusive form**: 0–10, 10–20, 20–30. The upper limit is **not** included, so 10 goes in 10–20.
- **Inclusive form**: 1–10, 11–20, 21–30. Both limits are included.
- **Class size (width)** = upper limit − lower limit (in exclusive form).
- **Class mark** $= \dfrac{\text{upper limit} + \text{lower limit}}{2}$.

Example 2.1 For the class 40–50, find the class size and the class mark.

SOLUTION

10 and 45

Example 2.2 The weights (in kg) of 15 students are: 32, 41, 38, 45, 36, 49, 43, 37, 40, 52, 34, 47, 39, 44, 50. Group them into classes 30–35, 35–40, 40–45, 45–50, 50–55.

SOLUTION

30–35: 32, 34. 35–40: 38, 36, 37, 39. 40–45: 41, 43, 40, 44. 45–50: 45, 49, 47. 50–55: 52, 50.

Frequencies 2, 4, 4, 3, 2

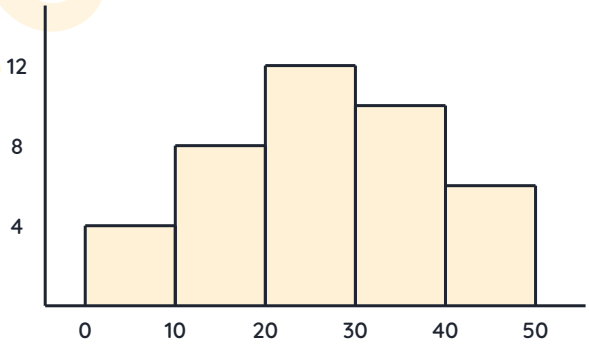
3 Bar graph

- A **bar graph** shows ungrouped data with bars of **equal width** and **equal gaps** between them. The height of a bar shows the value.
- Choose a suitable scale, label both axes and give the graph a title.
- A **double bar graph** shows two sets of data side by side for comparison.

4 Histogram

- A **histogram** shows **grouped continuous data**. Class intervals are on the horizontal axis and frequencies on the vertical axis.
- The bars have **no gaps**, because the classes are continuous.
- With equal class sizes, the heights of the bars are proportional to the frequencies.
- If the first class does not start at 0, a **kink** (zig-zag) is drawn on the horizontal axis.

Marks	0–10	10–20	20–30	30–40	40–50
Number of students	4	8	12	10	6



Histogram of the marks of 40 students

Example 4.1 From the histogram above: (i) which class has the highest frequency? (ii) How many students scored 30 or more? (iii) How many scored less than 20?

SOLUTION

(i) 20–30 (ii) $10 + 6 = 16$ **(iii)** $4 + 8 = 12$

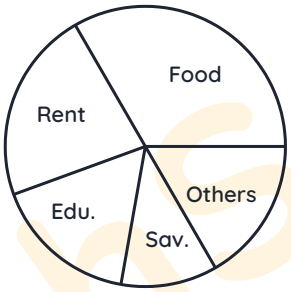
5 Pie chart

- A **pie chart** (circle graph) shows how a whole is divided into parts. Each part is a **sector**.
- The angle of each sector is proportional to the value it represents:

$$\text{Central angle} = \frac{\text{value of the component}}{\text{total value}} \times 360^\circ$$

- The central angles add up to 360°.
- To read a pie chart: value $= \frac{\text{central angle}}{360^\circ} \times$ total.

Item	Food	Rent	Education	Savings	Others	Total
Amount (₹)	6000	4000	3000	2000	3000	18000
Central angle	120°	80°	60°	40°	60°	360°



Pie chart of a family's monthly spending

Example 5.1 Find the central angles for the family budget in the table.

SOLUTION

Each ₹18,000 corresponds to 360°, so ₹1000 corresponds to 20°.

$120^\circ, 80^\circ, 60^\circ, 40^\circ, 60^\circ$

Example 5.2 In a pie chart of 720 students, the sector for "cricket" has an angle of 150°. How many students chose cricket?

SOLUTION

$\frac{150}{360} \times 720$

300

Example 5.3 A sector of 72° stands for ₹2400. Find the total amount shown by the pie chart.

SOLUTION

72° is $\frac{1}{5}$ of the circle.

₹12,000

Example 5.4 In a pie chart, 35% of the students walk to school. Find the angle of that sector.

SOLUTION

$0.35 \times 360^\circ$

126°

6 Choosing a graph

Data	Suitable graph
Separate items to compare	Bar graph
Two related sets to compare	Double bar graph
Grouped continuous data	Histogram
Parts of a whole	Pie chart

7 Common mistakes

- Putting an observation equal to an upper limit in that class (exclusive form). It belongs to the next class.
- Leaving gaps between the bars of a histogram, or drawing a bar graph without gaps.
- Central angles that do not add up to 360°.
- Using percentages as angles: 25% is 90°, not 25°.
- Forgetting to check that the frequencies add up to the number of observations.
- Not stating the scale.

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